Postlab report 3 - DSP

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Lab 3: Sampling and reconstruction

1) Aliasing error

1. We connected and adjusted the system according to the diagram in the report :

ציור 1.2 : מערכת הדגימה

Figure 1: Diagram of theoretical sampling system.
We build the same system with the lab equipment.

2. We are asked to create a square signal and sample it with the following characteristics:

   - sample time - 1 sec
   - sample rate - 3200khz
   - signal base frequency - 250hz
   - signal amplitude - based on equipment accuracy range...

3. We sample the signal at said rate, collect the data samples using matlab and apply an FFT for it.
   We reconstruct the signal for 2 separate cases :

(a) **Reconstruct the signal as it is**

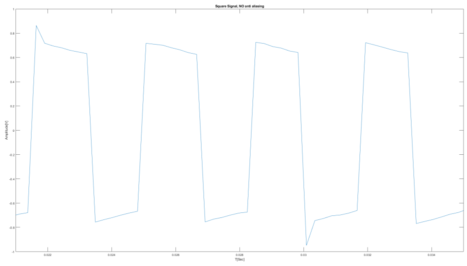

Figure 2: Square wave in time domain - 250hz sampled in rate 3200hz
Reconstructed without anti-aliasing filter

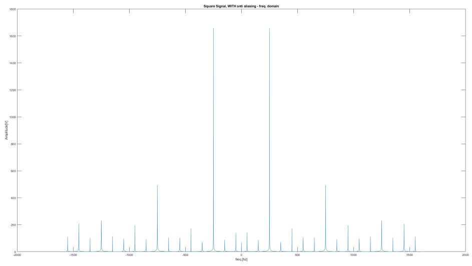

Figure 3: Square wave in freq. domain - 250hz sampled in rate 3200hz
Reconstructed without anti-aliasing filter

We can see that the frequency spectrum of the reconstructed signal contains harmonies that are theoretically not a part of a square signal.
That is due to aliasing created form yielded duplicates of the spectrum do to the sampling process.

(b) **Apply anti-aliasing filter before reconstructing**

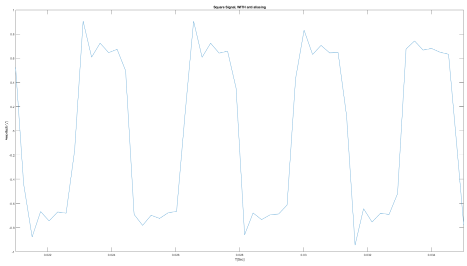

Figure 4: Square wave in freq. domain - 250hz sampled in rate 3200hz Reconstructed with anti-aliasing filter with cutoff freq. of 1600hz

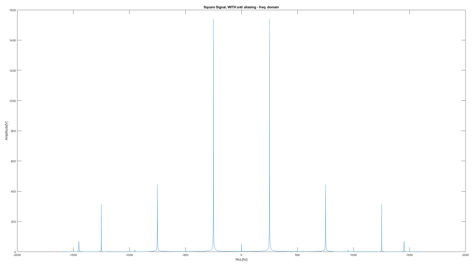

Figure 5: Square wave in freq. domain - 250hz sampled in rate 3200hz Reconstructed with anti-aliasing filter with cutoff freq. of 1600hz

We can barely see that the frequency spectrum of the reconstructed signal contains the same harmonies as before but they are dimished greatly.
That is due to the anti-aliasing filter we applied before reconstruction.

4. We can see the effectiveness of the anti-aliasing filter.
Without it, we are left with non-negligible, residue harmonies in the spectrum of the reconstructed signal.

5. We are asked to zero-padd the signals in the frequency domain and then reconstruct them back to time domain. This operation simulates sampling the signal at a higher rate than the original.
   But in our case, we are trying to simulate the higher sampling rate of a band-limited signal with a rate that is higher than its bandwidth. In practice that means we are trying to get more data then what's stored in the signal. Since that is not possible, the added samples are actually the interpolation with the sinc function that represents the bandwidth window in time domain.

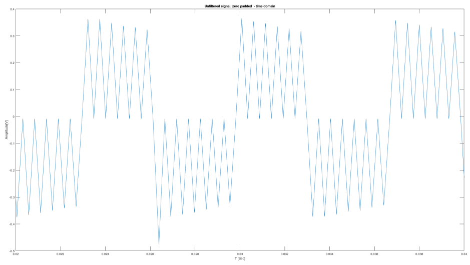

Figure 6: Unfiltered square wave reconstructed

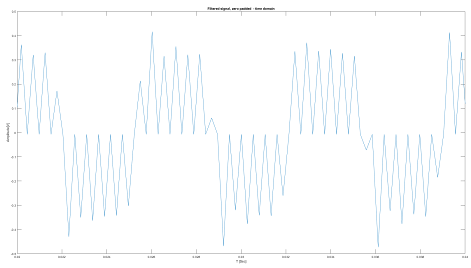

Figure 7: filtered square wave reconstructed

2) Quantization error

1. We are asked to sample a triangular wave with the following properties :

   - Base frequency : $f = 250hz$
   - Number of samples : $L_0 = 16,494 \geq 3000$

   The rest of the system characteristics are upto the user. I chose them as follows :

   - L_0 : I chose the specific number of samples arbitrarily But what i did chose is the sampling rate and time such that the number of samples was derived from it. When sampling we need to make sure that the number of samples is indeed high enough to resemble a proper distribution. We also need to make sure that the sampling rate does shares a minimal number to no common dividers so we won't sample at the same relative amplitudes every time period.
   - Amplitude : We don't want to lose any information beforehand so we make sure that the presampled signal is within the range of the sampler to prevent any truncation error.

2. We simulate the passage of the signal inside a quantizer with different levels of accuracy. We look at different numbers of representing bits and examine the distribution of the quantization error. Here are the results :

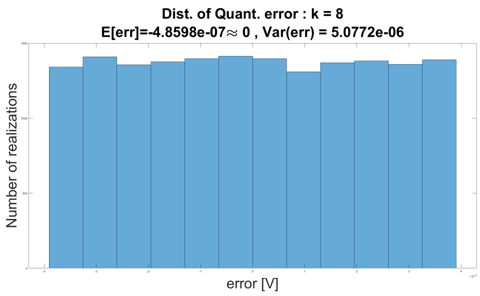

Figure 8: Distribution of 8-bit quantization

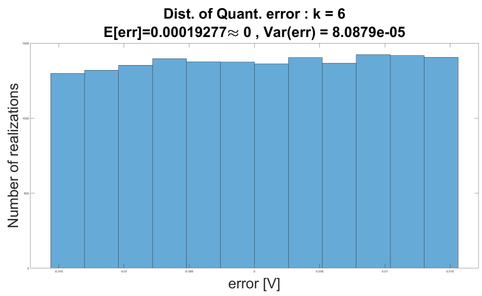

Figure 9: Distribution of 6-bit quantization

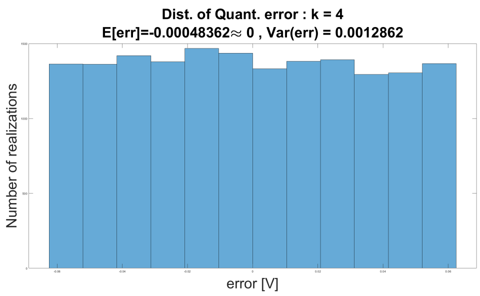

Figure 10: Distribution of 4-bit quantization

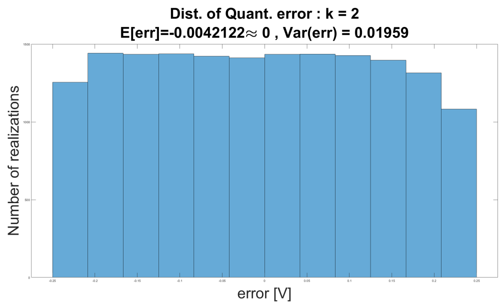

Figure 11: Distribution of 2-bit quantization

   A few key observations :

   - All distributions show near uniform distribution : $E_q := X - X_q \sim U[-\frac{\Delta}{2}, \frac{\Delta}{2}]$
   - All distributions show near zero expected value : $\mathbb{E}[E_q] = 0$
   - All distributions show variance value : $Var(E_q) = \frac{\Delta^2}{12}$

3) Reconstruction

1. We are asked to sample a sine wave with optimal sampling conditions so we set the following conditions :

   - Base freq. : $f = 250hz$
   - Amplitude (within the dynamic range of the sampler) : $A = 500mV_{pp}$
   - Sampling rate (satisfying Nyquist) : $f_s = 600hz > 2 \cdot f$

2. We will **not use an anti-aliasing filter** since we want to demostrate the consequences of violating Nyquists condition.

3. We connected and adjusted the system according to the diagram in the report :

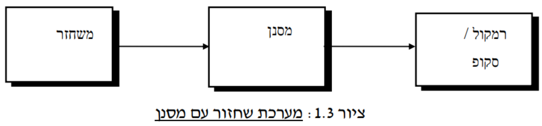

Figure 12: Diagram of theoretical sampling system.
We build the same system with the lab equipment.

4. We gradually raise the freq. of the sine wave and reconstruct for each case while keeping the same sampling rate. here are the results :

   Here are the recorded signals :

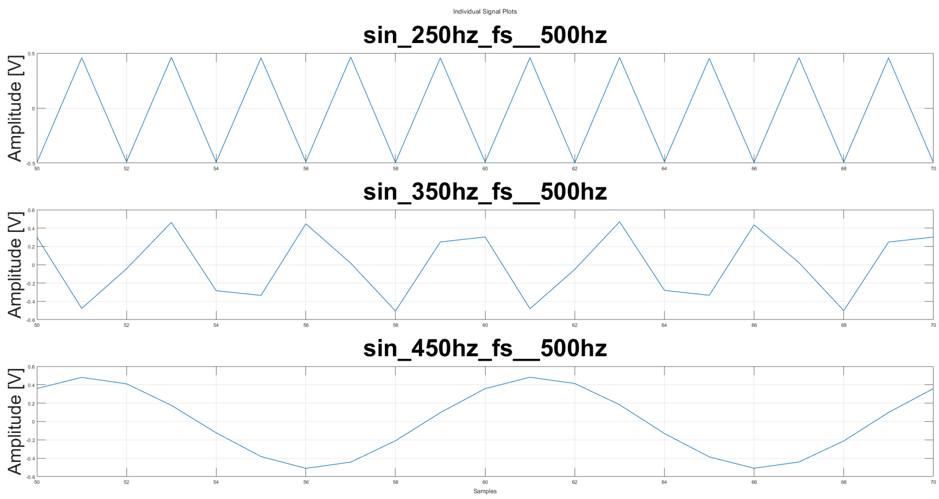

Figure 13: signals with different frequencies sampled at the same sampling rate

And here are the reconstructions of the recorded signals :

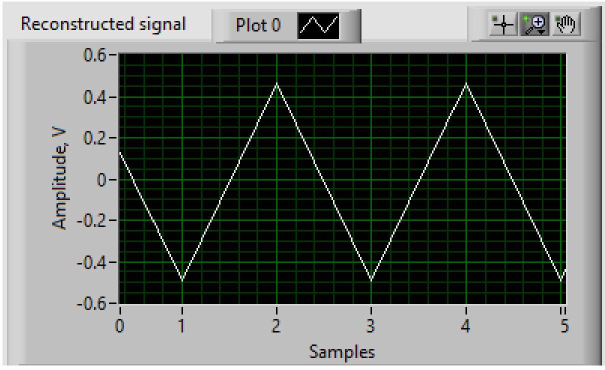

Figure 14: reconstruction of $f = 250hz$

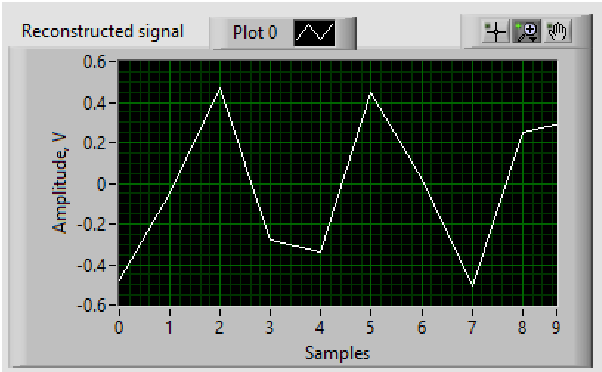

Figure 15: reconstruction of $f = 350hz$

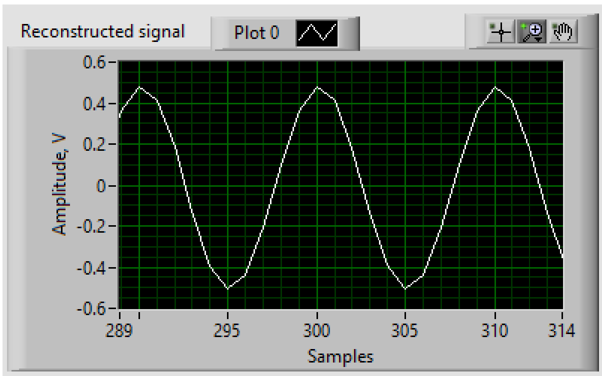

Figure 16: reconstruction of $f = 450hz$

5. We can see that only the 250hz sine wave was reconstructed correctly while the rest either don't resemble the original signal or reconstructed in a different frequency. That illustrates the importance of Nyquists condition.

4) Voice Recording and Reconstruction

1. We are asked to record our own voice saying in hebrew : "apocalypse now"

   - Recording time : $T_s = 2Sec$
   - Sampling rate : $f_s = 8khz$ - We set the sampling rate assuming that speech freq. is nowhere higher than $f_{speech} \approx 4khz$
   - The rest is set by the system...

2. We perform spectrum estimation using Welch's estimation to extract the PSD of the speech signal :

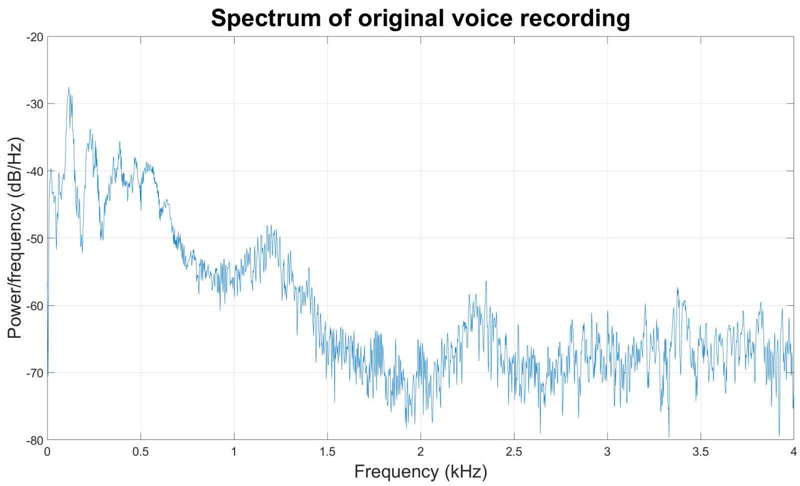

Figure 17: Welch's spectrum estimation of vocal recording

3. It is very visible that most of the signal's energy is in **low frequencies**, so we might be able to record in a lower sampling rate.

4. We reconstruct the signal in time domain, both in plotting and playing the actual sound in voice recording using matlabs "soundsc()" function:

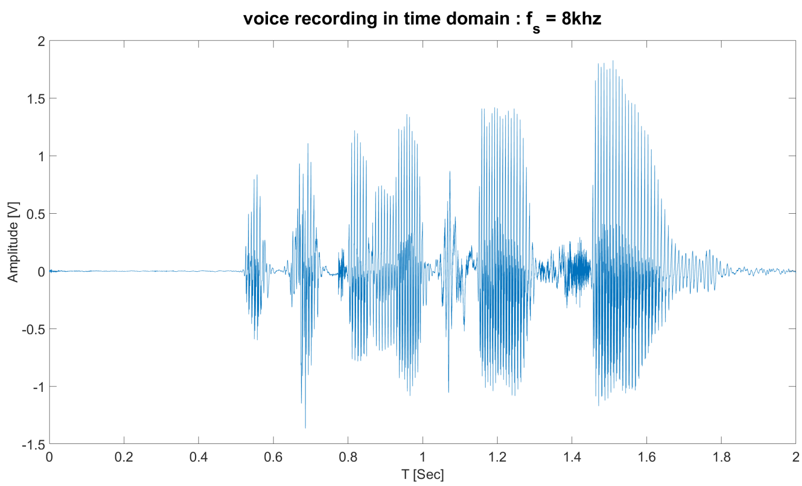

Figure 18: Vocal recording in time domain

5. We are asked to simulate the sampling of the signal using different resolution sampling card. We simulate for resolution : 12bit, 8bit, 4bit, 2bit quantization.

6. We perform the recostruction of the signals and plot them against each other to visualize the difference :

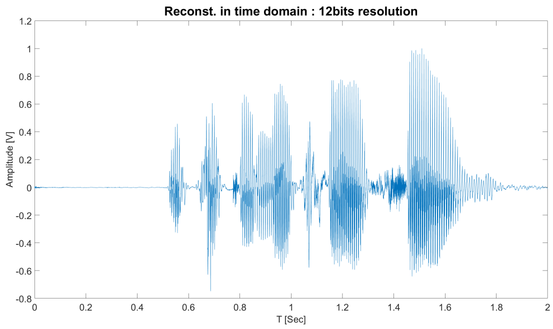

Figure 19: Reconstruction with 8-bit resolution

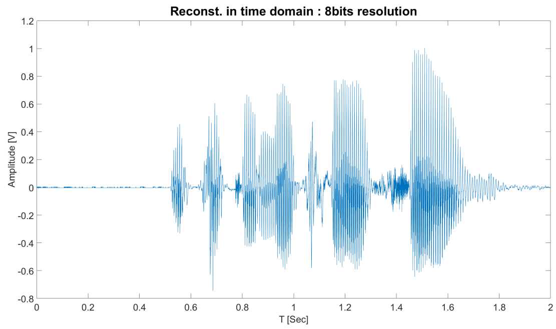

Figure 20: Reconstruction with 6-bit resolution

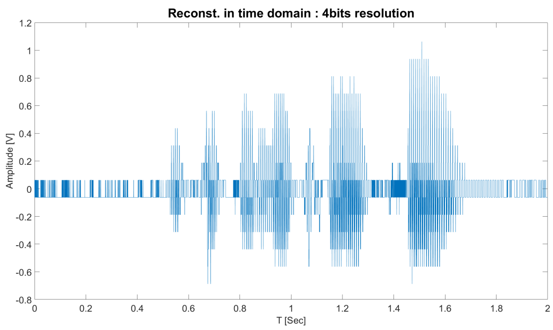

Figure 21: Reconstruction with 4-bit resolution

Figure 22: Reconstruction with 2-bit resolution

After listening to the recordings :

- For the 12bit resolution the signal still sounded quite well. There was no noticeable distortion or decrease in quality.
- For the 8bit and 4bit resolution the signal quality started to decline rapidly, the sound was more distorted and "metalic" though still quite understandable even for the 4bit...
- For the 2bit resolution there was a sharp decline in the signal quality and it would be hard to guess what was said in the recording previously.

7. We are Now asked to assume that the signals band width is $BW = 3.2khz$ (since it was received through a phone line...) . We are asked to design a new sampling system with resolution of 8bit and sample a new recording using it.

   For that we will :

   - First sample the signal at $6.4khz$.
   - Then use an anti-aliasing with cutoff $3.2khz$ to diminish any excess frequencies and satisfy Nyquist's condition.
   - Then simulate the 8bit quantization last.

   Here are the results :

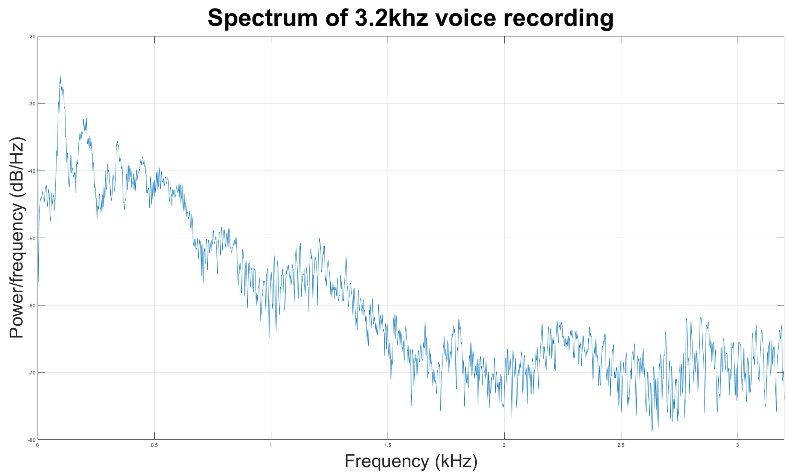

Figure 23: Welch's spectrum estimation of 3.2khz band-limited vocal recording

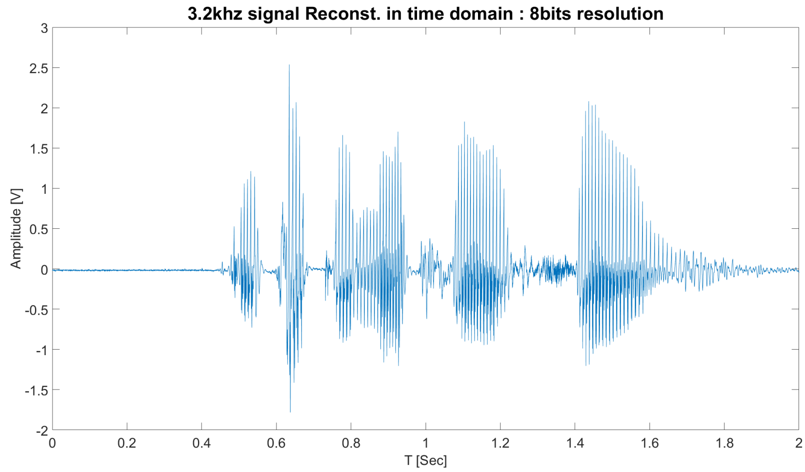

Figure 24: 3.2khz band-limited vocal recording in time domain

   Key observations:

   - As we hypothesized in 4.3, most of the signals energy remains in the bandwidth of the new signal so we can safely say that most of the information is saved.
   - Visually, the signal seems similar to the original recording so that's reassuring.

- When listening to the sound recording, the quality is fairly high and the sound is understandable. We can see that even under the limitations forced on us we could still communicate through the channel!